Vedha Mahesh

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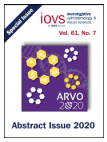
EDUCATION

Gunn High School (Palo Alto, CA)
GPA: **3.7** Weighted GPA**: 3.9** ACT: **34/36** Super Score: **35/36**
SAT Biology **(780/800)** SAT Math2 **(800/800)**
Awards: President's Voluntary Service award, **Girl Scouts Gold Award,** AP Scholar with Distinction
Skills: Python, Java, CSS, HTML, SQL for Data Science
Memberships: DECA Business, Best Buddies, Cross Country

SCIENTIFIC PUBLICATIONS

Comparison of visual function and retinal structure between phases of multiple sclerosis
Jennifer Yarp; Fareshta Khushzad; Lakshmi Leishangthem; Vedha Mahesh; Anna Tomczak; May Han; Lucas Kipp; Heather Moss
Investigative Ophthalmology & Visual Science June 2020, Vol.61, 5103. doi: **Contribution**: I analyzed over 500 OCT scans (Avanti, Optovue) from patients with multiple sclerosis (MS) and recorded various measurements from those scans including the thickness of the macula ganglion cell complex(GCC) and peripapillary retina nerve fiber layer(RNFL)



Characterization of retinal vascular changes in multiple sclerosis using adaptive optics and OCTA
Fareshta Khushzad; Jennifer Yarp; Aubrey Hargrave; Nripun Sredar; Vedha Mahesh; Anna Tomczak; Lucas Kipp; May Han; Alfredo Dubra; Heather Moss
Investigative Ophthalmology & Visual Science June 2020, Vol.61, 5104. doi: **Contribution:** I examined over 400 OCTA and AO-SLO images of 11 MS subjects with RRMS and SPMS and recorded data including measurements of the RVD around the optic nerve and in superficial and deep layers of the macula

EXPERIENCE

Stanford Byers Eye Institute, Palo Alto *(Research Intern)* Jun 2019 – Mar 2020
- Worked with Dr. Moss on multiple projects focusing on diseases of the optic nerve
- Analyzed MRI images to help develop patient specific models of eye geometry in high intracranial pressure states
- Submitted research on visual function and retinal structure in subjects with multiple sclerosis to NANOS and ARVO

UCSF Okada Lab, San Francisco *(Research Lab Student)* Dec 2018 – Dec 2019
- Worked under a cellular immunologist studying T-cell therapies for glioma immunotherapy
- Learned skills such as PCR, gel electrophoresis, and plasmid DNA isolation

Bio-Informatics Workshop, Berkeley *(Trainee)* Dec 2018
- Performed gene expression profiling and analysis of sequence variation
- Analyzed large, publicly available data sets

Abilities United, Palo Alto *(Volunteer)* Feb 2018 – Feb 2020
- Served as a summer camp and afterschool program counselor to serve children with developmental disabilities
- Coordinated fun activities and field trips to museums, parks, and zoos for children ages 5-16
- Helped plan the logistics for trips to the Happy Hollows Zoo and Palo Alto Baylands Park

Sanskruti Shikar (School for Special Needs Children), Hyderabad *(Project Lead)* Jun 2017 – Aug 2017
- Presented solutions about better sleeping, eating, and exercising habits to the Principal and the Board
- Raised over $5,500 from multiple fundraising events to help the program with bunk beds, and exercise equipment
- Designed and implemented a new health and wellness curriculum and spread awareness about rising problem of obesity in children with special needs

JEI (Local Tutoring Center), La Cañada *(Paid Tutor)* Jun 2017 – Aug 2017
- Tutored and graded papers for elementary school kids in Math at JEI

LEADERSHIP

Girl Scouts *(Leadership Board Member)* Sep 2017 – Present
- Led monthly meetings and organized service projects for our troop and younger troops
- Coordinated with Haight Ashbury Food Program's food pantry to distribute groceries to families in need
- Generated campaigns for various fund-raising activities; Successfully launched 6 unique campaigns, accruing over $5,500 for Silver and Gold Awards

Indian Culture Club *(Co-President/Co-Founder)* Aug 2019 – Present
- Hosted weekly meetings with activities, food, and other elements to showcase and educate about Indian Culture

La Cañada Elementary Tech Night *(STEM evangelist)* Jun 2017 – Aug 2017
- Hosted multiple "Tech Nights" for elementary school students to introduce them to robotics through Dash & Dot and Brushbots (DIY toothbrush robot), and taught basic coding principles and skills using Arduino
- Encouraged participation from girls through a Princess Coding Night event